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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

SALARY PAYROLL PROCESSING AND PAYSLIP GENERATION SYSTEM

Submitted by

HARIHARAN D

25BCS115

Under the Guidance of

Mr. ARAVIND

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled "Salary Payroll Processing and Payslip Generation System" submitted in partial fulfillment of the requirements for the course [JAVA PROGRAMMING] is a record of my original work carried out under the guidance of [Mr.ARAVIND], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HARIHARANN D

(Your Name)

CERTIFICATE

This is to certify that the project entitled "Salary Payroll Processing and Payslip Generation System" is a bona fide work carried out by [HARIHARAN D] [I-' B'], [BSC. COMPUTER SCIENCE], KG College of Arts & Science, in partial fulfillment of the requirements for the course [JAVA PROGRAMMING].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place:COIMBATORE

Date:10.04.2026

Submitted for the Viva-Voce Examination held on _____

Guide	Head of the Department
(Technical Trainer, KGCAS)	(Technical Head, KGCAS)
Mr.Aravind	Mr.Boopalan sundaramoorthy

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ABSTRACT

The Salary Payroll Processing and Payslip Generation System is a Java-based application designed to automate the complete payroll management process in organizations. In many small and medium-sized enterprises, payroll calculations are performed manually, which often results in errors, delays in salary disbursement, and inefficient record maintenance. This project aims to simplify and automate payroll operations by providing a systematic digital solution.

The system allows HR personnel or administrators to enter employee details, define salary components such as basic pay, HRA, DA, PF, and other allowances or deductions, and automatically compute the net salary payable to each employee. A professional payslip is generated displaying all salary components clearly.

The application is developed using Core Java programming concepts, including classes, objects, methods, inheritance, encapsulation, loops, conditional statements, and exception handling. The system can be implemented as a console-based application or enhanced with Java Swing for a graphical user interface.

The primary objective of this system is to improve payroll accuracy, reduce processing time, and generate professional payslips for employees. This project demonstrates the practical application of object-oriented programming concepts in Java for real-world business automation.

1. INTRODUCTION

In any organization, whether a corporate enterprise, educational institution, or small business, managing employee salaries is one of the most critical administrative responsibilities. Traditionally, payroll management has been done manually using registers, spreadsheets, or handwritten records. This approach is prone to calculation errors, time-consuming, and difficult to scale as the number of employees grows.

With the rapid advancement of computer technology, automated payroll systems have become essential tools for organizations to manage salary processing efficiently and accurately. An automated payroll system reduces manual effort, minimizes errors, ensures timely salary payments, and generates professional payslips for employees.

The Salary Payroll Processing and Payslip Generation System is a Java-based application designed to automate the end-to-end process of employee salary computation and payslip generation. The system accepts employee details such as name, employee ID, department, designation, and salary components including basic pay, House Rent Allowance (HRA), Dearness Allowance (DA), and deductions such as Provident Fund (PF) and Professional Tax (PT). It then computes the gross salary, total deductions, and net salary automatically.

The system generates a structured payslip displaying all salary components clearly, which can be viewed or printed for distribution to employees. This project is developed using Core Java concepts such as classes, objects, methods, inheritance, encapsulation, loops, and exception handling.

1.1 Objective

The main objective of the Salary Payroll Processing and Payslip Generation System is to develop an efficient Java-based application that automates payroll calculations and generates employee payslips accurately and quickly.

The specific objectives of this project are:

- To design and develop a Java-based payroll processing system.
- To automate salary computation including allowances and deductions.
- To generate professional and structured payslips for employees.
- To reduce manual errors in payroll calculations.
- To maintain employee salary records systematically.
- To demonstrate Core Java programming concepts such as classes, objects, inheritance, and encapsulation.
- To provide a user-friendly interface for HR personnel to manage payroll operations.

1.2 Overview

The Salary Payroll Processing and Payslip Generation System is a menu-driven Java application that enables HR

administrators to manage employee payroll efficiently. The system provides a centralized platform where employee salary information can be entered, processed, and displayed in the form of a payslip.

The application allows users to enter employee details and salary components. It then computes the gross salary by summing up basic pay and all allowances, calculates total deductions including PF, professional tax, and other deductions, and derives the net salary payable to the employee.

A structured payslip is generated displaying the employee information, salary breakup, deductions summary, and final net pay. The system can manage multiple employees and process payroll for a selected month.

1.3 Features

The Salary Payroll Processing and Payslip Generation System includes the following key features:

- Employee Management – Add, view, and manage employee records including name, ID, department, and designation.
- Salary Component Configuration – Define and manage salary components such as Basic Pay, HRA, DA, TA, and Special Allowances.
- Automatic Salary Calculation – Automatically compute gross salary, total deductions, and net salary.
- Deduction Management – Calculate and apply deductions such as Provident Fund (PF), Professional Tax (PT), and other deductions.
- Payslip Generation – Generate a structured and professional payslip for each employee.
- Multiple Employee Processing – Process payroll for multiple employees in a single session.
- Input Validation – Validate user inputs to prevent incorrect data entry.
- User-Friendly Interface – Provide a menu-driven interface for easy navigation.

2. PROGRAM ANALYSIS

Program analysis is a fundamental phase in software development that involves studying the problem in depth and identifying the requirements of the system. This stage helps in understanding the existing challenges and proposing an automated solution that addresses those issues effectively.

2.1 Problem Definition

In many organizations, especially small and medium-sized enterprises, employee payroll is managed manually using spreadsheets or handwritten records. This approach creates several challenges that affect payroll accuracy and efficiency.

Common problems in manual payroll systems include:

- Errors in salary calculation due to manual arithmetic.
- Time-consuming process for computing multiple employee salaries.
- Difficulty in maintaining and retrieving salary records.
- Inconsistency in applying allowances and deductions across employees.
- Delays in salary disbursement due to slow processing.
- Lack of standardized payslip formats for employees.

These problems lead to employee dissatisfaction, compliance risks, and administrative inefficiency. Therefore, an automated Salary Payroll Processing and Payslip Generation System is required to address these challenges.

2.2 Project Overview

The Salary Payroll Processing and Payslip Generation System is a Java-based desktop application that automates the payroll processing workflow in an organization. The system manages employee data, processes salary components, and generates payslips automatically.

The system handles the following salary components:

Component	Type	Calculation Method
Basic Pay	Earning	Fixed amount entered by administrator
House Rent Allowance (HRA)	Earning	40% of Basic Pay
Dearness Allowance (DA)	Earning	10% of Basic Pay
Travel Allowance (TA)	Earning	Fixed amount or 5% of Basic Pay
Special Allowance	Earning	Fixed amount as applicable

Provident Fund (PF)	Deduction	12% of Basic Pay
Professional Tax (PT)	Deduction	Fixed slab-based deduction
Income Tax (TDS)	Deduction	As per applicable tax slab

The system computes gross salary as the sum of all earnings, total deductions as the sum of all deductions, and net salary as gross salary minus total deductions.

2.3 Scopes and Limitations

Scope

- The system can manage payroll for multiple employees within a single session.
- Salary components including allowances and deductions are configurable.
- Payslips can be generated and displayed for individual employees.
- The system supports payroll processing for monthly salary cycles.
- Employee data can be entered and modified during runtime.

Limitations

- The system does not include database connectivity for persistent data storage in the basic version.
- The application does not support multi-user access or role-based permissions in the basic implementation.
- Payslip printing requires system printer configuration; direct print functionality is not built-in.
- Tax calculations are based on simplified slab rates and may not cover all complex tax scenarios.
- The basic version does not support payroll history across multiple months.

3. SYSTEM DESIGN

System design describes the overall architecture and structure of the Salary Payroll Processing and Payslip Generation System. The system is organized into well-defined modules, each responsible for a specific functionality within the payroll process.

3.1 Modules

1. Employee Management Module

This module handles all operations related to employee data.

Key functions include:

- Accepting employee details such as Employee ID, Employee Name, Department, and Designation.
- Storing employee records for payroll processing.
- Displaying employee information as part of the payslip.

2. Salary Component Module

This module manages the salary structure for each employee.

Key functions include:

- Accepting basic pay input from the administrator.
- Calculating HRA as 40% of Basic Pay.
- Calculating DA as 10% of Basic Pay.
- Calculating TA as 5% of Basic Pay.
- Accepting or computing Special Allowance.

3. Deduction Calculation Module

This module computes all salary deductions applicable to the employee.

Key functions include:

- Calculating Provident Fund (PF) as 12% of Basic Pay.
- Calculating Professional Tax (PT) based on salary slab.
- Computing total deductions.

4. Net Salary Computation Module

This module calculates the final net salary payable to the employee.

Computation logic:

- Gross Salary = Basic Pay + HRA + DA + TA + Special Allowance
- Total Deductions = PF + Professional Tax + Other Deductions
- Net Salary = Gross Salary − Total Deductions

5. Payslip Generation Module

This module generates and displays the structured payslip for the employee.

The payslip includes:

- Employee information (Name, ID, Department, Designation, Month)
- Earnings breakdown (Basic Pay, HRA, DA, TA, Special Allowance, Gross Salary)
- Deductions breakdown (PF, PT, Total Deductions)
- Net salary payable

6. Exit Module

This module provides a safe exit option for the user after completing payroll processing operations. The system terminates gracefully when the user selects the exit option.

4. METHODOLOGY

The methodology describes the structured approach followed to develop the Salary Payroll Processing and Payslip Generation System. A step-by-step software development process was followed to ensure the system is well-designed, functional, and thoroughly tested.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problems associated with manual payroll processing were studied and analyzed in detail. The requirements for an automated payroll system were identified based on the needs of HR departments in organizations.

Key problems identified include:

- Manual salary calculations are error-prone and time-consuming.
- Inconsistent application of allowances and deductions across employees.
- Difficulty in maintaining payroll records for multiple employees.
- Absence of standardized payslip formats.

Based on this analysis, the system was designed to automate salary computation, apply allowances and deductions systematically, and generate structured payslips automatically.

4.2. System Design

In this phase, the overall structure of the system was designed before coding began. The system was divided into six modules as described in Chapter 3.

The system architecture was designed using Object-Oriented Programming principles:

- An Employee class was designed to store employee information.
- A Salary class was designed to manage salary components.
- A PayrollProcessor class was designed to handle computation logic.
- A PayslipGenerator class was designed to format and display payslips.

4.3. Program Development

The program was implemented using the Java programming language following the modular design defined in the system design phase.

Development steps followed:

- Creation of Java classes for Employee, Salary, Deductions, and PayslipGenerator.
- Implementation of salary calculation methods.
- Implementation of deduction calculation logic.
- Writing the main menu-driven program for user interaction.
- Implementing payslip formatting and display logic.

Core Java concepts applied during development:

- Classes and Objects – Representing employees and salary components.
- Encapsulation – Using private fields with getter and setter methods.
- Inheritance – Extending base classes for different employee types.
- Methods – Modular functions for computation and display.
- Loops and Conditional Statements – Menu-driven logic and salary calculations.
- Exception Handling – Managing invalid user inputs.

4.4. Testing and Validation

Testing was performed to ensure the system functions correctly and produces accurate results. Multiple test cases were executed to validate different scenarios.

Test Case	Input	Expected Result	Status
Enter employee details	Valid name and ID	Employee record created successfully	Pass
Enter basic pay	50000	Correct salary components calculated	Pass
HRA Calculation	Basic = 50000	HRA = 20000 (40%)	Pass
PF Deduction	Basic = 50000	PF = 6000 (12%)	Pass
Net Salary	Gross=70000, Ded=8000	Net = 62000	Pass
Invalid input	Negative salary value	Error message displayed	Pass
Payslip generation	Valid employee data	Structured payslip displayed	Pass

4.5. Result and Documentation

After successful testing, the system results were documented. The program successfully processes payroll for multiple employees and generates accurate payslips displaying all salary components clearly.

The results demonstrate that the system reduces manual effort, eliminates calculation errors, and produces professional payslips in a structured format. Proper documentation was prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Salary Payroll Processing and Payslip Generation System:

1. Start the program.
2. Display the main menu with options: Add Employee, Process Payroll, Generate Payslip, Exit.
3. If user selects Add Employee: accept Employee ID, Name, Department, and Designation.
4. If user selects Process Payroll: accept Basic Pay for the employee.
5. Calculate HRA = 40% of Basic Pay.
6. Calculate DA = 10% of Basic Pay.
7. Calculate TA = 5% of Basic Pay.
8. Calculate Gross Salary = Basic + HRA + DA + TA + Special Allowance.
9. Calculate PF Deduction = 12% of Basic Pay.
10. Calculate Professional Tax based on salary slab.
11. Calculate Total Deductions = PF + PT + Other Deductions.
12. Calculate Net Salary = Gross Salary − Total Deductions.
13. If user selects Generate Payslip: display formatted payslip with all components.
14. If user selects Exit: terminate the program.
15. End the program.

4.7. Flowchart

The flowchart below represents the program flow of the Salary Payroll Processing and Payslip Generation System:

Start

↓

Display Main Menu

↓

Select Option: [1] Add Employee [2] Process Payroll [3] Generate Payslip [4] Exit

↓

Option 1 → Enter Employee Details → Store Record

Option 2 → Enter Basic Pay → Calculate HRA, DA, TA, PF, PT → Compute Gross & Net Salary

Option 3 → Display Payslip with all Components

Option 4 → Exit Program

↓

End

Students can draw this flowchart using draw.io, Lucidchart, or MS Visio and insert the image here.

5. SOURCE CODE

Below is the complete Java source code for the Salary Payroll Processing and Payslip Generation System.

```
// ===== Employee.java =====
public class Employee {
    private int empId;
    private String name, department, designation;

    public Employee(int empId, String name, String department, String designation) {
        this.empId = empId;
        this.name = name;
        this.department = department;
        this.designation = designation;
    }
    // Getters
    public int getEmpId() { return empId; }
    public String getName() { return name; }
    public String getDepartment() { return department; }
    public String getDesignation() { return designation; }
}

// ===== Salary.java =====
public class Salary {
    private double basicPay;
    private double hra, da, ta, specialAllowance;
    private double pf, professionalTax;

    public Salary(double basicPay, double specialAllowance) {
        this.basicPay = basicPay;
        this.specialAllowance = specialAllowance;
        this.hra = basicPay * 0.40;
        this.da  = basicPay * 0.10;
        this.ta  = basicPay * 0.05;
        this.pf  = basicPay * 0.12;
        this.professionalTax = calculatePT(basicPay);
    }

    private double calculatePT(double basic) {
        if (basic <= 10000) return 0;
        else if (basic <= 20000) return 150;
        else return 200;
    }

    public double getGrossSalary() {
        return basicPay + hra + da + ta + specialAllowance;
    }
    public double getTotalDeductions() {
        return pf + professionalTax;
    }
    public double getNetSalary() {
        return getGrossSalary() - getTotalDeductions();
    }
    // Getters for all fields...
    public double getBasicPay() { return basicPay; }
    public double getHra() { return hra; }
    public double getDa() { return da; }
    public double getTa() { return ta; }
    public double getPf() { return pf; }
```

```
    public double getProfessionalTax() { return professionalTax; }
}

// ===== PayrollMain.java (Main Class) =====
import java.util.Scanner;

public class PayrollMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        Employee emp = null;
        Salary salary = null;
        do {
            System.out.println("\n===== PAYROLL SYSTEM =====");
            System.out.println("1. Add Employee");
            System.out.println("2. Process Payroll");
            System.out.println("3. Generate Payslip");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt(); sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Employee ID: "); int id = sc.nextInt(); sc.nextLine();
                    System.out.print("Name: "); String name = sc.nextLine();
                    System.out.print("Department: "); String dept = sc.nextLine();
                    System.out.print("Designation: "); String desig = sc.nextLine();
                    emp = new Employee(id, name, dept, desig);
                    System.out.println("Employee added successfully!");
                    break;
                case 2:
                    System.out.print("Basic Pay: "); double basic = sc.nextDouble();
                    System.out.print("Special    Allowance:    ");    double    special    =
sc.nextDouble(); sc.nextLine();
                    salary = new Salary(basic, special);
                    System.out.println("Payroll processed!");
                    break;
                case 3:
                    generatePayslip(emp, salary);
                    break;
                case 4: System.out.println("Exiting..."); break;
                default: System.out.println("Invalid choice!");
            }
        } while (choice != 4);
    }

    static void generatePayslip(Employee e, Salary s) {
        System.out.println("\n========== PAYSLIP ==========");
        System.out.println("Emp ID    : " + e.getEmpId());
        System.out.println("Name      : " + e.getName());
        System.out.println("Department: " + e.getDepartment());
        System.out.println("Designation: " + e.getDesignation());
        System.out.println("-----------------------------");
        System.out.println("EARNINGS");
        System.out.printf("Basic Pay        : Rs. %.2f%n", s.getBasicPay());
        System.out.printf("HRA (40%%)        : Rs. %.2f%n", s.getHra());
        System.out.printf("DA (10%%)         : Rs. %.2f%n", s.getDa());
        System.out.printf("TA (5%%)          : Rs. %.2f%n", s.getTa());
        System.out.printf("Gross Salary     : Rs. %.2f%n", s.getGrossSalary());
        System.out.println("-----------------------------");
        System.out.println("DEDUCTIONS");
        System.out.printf("PF (12%%)         : Rs. %.2f%n", s.getPf());
        System.out.printf("Professional Tax  : Rs. %.2f%n", s.getProfessionalTax());
        System.out.printf("Total Deductions : Rs. %.2f%n", s.getTotalDeductions());
```

```
        System.out.println("==============================");
        System.out.printf("NET SALARY       : Rs. %.2f%n", s.getNetSalary());
        System.out.println("Thank You!");
    }
}
```

6. INPUT AND OUTPUT SCREENS

The following screens represent the input and output of the Salary Payroll Processing and Payslip Generation System during program execution.

Figure 1: Main Menu Screen

```
===== PAYROLL SYSTEM =====
1. Add Employee
2. Process Payroll
3. Generate Payslip
4. Exit
Enter choice: 1
```

Figure 2: Employee Details Entry

```
Employee ID: 1001
Name: Rajesh Kumar
Department: IT
Designation: Software Engineer
Employee added successfully!
```

Figure 3: Payroll Processing Input

```
Basic Pay: 50000
Special Allowance: 5000
Payroll processed!
```

Figure 4: Generated Payslip Output

```
========== PAYSLIP ==========
Emp ID     : 1001
Name       : Rajesh Kumar
Department: IT
Designation: Software Engineer
----------------------------
EARNINGS
Basic Pay          : Rs. 50000.00
HRA (40%)          : Rs. 20000.00
DA (10%)           : Rs. 5000.00
TA (5%)            : Rs. 2500.00
Special Allowance: Rs. 5000.00
Gross Salary       : Rs. 82500.00
----------------------------
DEDUCTIONS
PF (12%)           : Rs. 6000.00
Professional Tax   : Rs. 200.00
Total Deductions : Rs. 6200.00
============================
NET SALARY         : Rs. 76300.00
Thank You!
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Salary Payroll Processing and Payslip Generation System was developed to automate the complete payroll management process in organizations. The system successfully demonstrates how Core Java programming concepts can be applied to solve real-world business problems in human resource management.

During the development of this project, key Java programming concepts such as classes, objects, encapsulation, inheritance, methods, loops, conditional statements, and exception handling were applied effectively. The system was structured into well-defined modules including Employee Management, Salary Computation, Deduction Calculation, and Payslip Generation, ensuring clean code organization and maintainability.

The salary calculation logic correctly implements standard payroll rules, computing HRA at 40% of basic pay, DA at 10%, TA at 5%, and PF deduction at 12%. Professional Tax is calculated based on salary slabs, reflecting standard payroll practices. The system was tested with multiple test cases covering valid inputs, edge cases, and invalid inputs, all of which produced expected results.

The payslip generated by the system provides a clear, structured breakdown of earnings and deductions, making it easy for employees to understand their salary components. The system performs all calculations automatically, eliminating manual errors and significantly reducing processing time compared to traditional manual payroll methods.

7.2 Conclusion

The Salary Payroll Processing and Payslip Generation System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world business automation application.

The system effectively automates salary computation by calculating all standard allowances and deductions, and generates professional payslips for employees. It reduces manual effort, minimizes calculation errors, and ensures accurate salary processing. The modular design of the system makes it easy to understand, maintain, and extend with additional features.

This project helped in developing a strong understanding of object-oriented programming principles in Java, including the design and implementation of classes, methods, inheritance, and encapsulation in a practical business context.

The system can be further enhanced in the future by adding features such as database connectivity using JDBC for persistent data storage, a graphical user interface using Java Swing or JavaFX, multi-employee batch payroll processing, payroll history across multiple months, PDF payslip generation and email distribution, and

integration with an employee attendance management system.

8. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – Java: The Complete Reference, 11th Edition, McGraw-Hill Education.
- Cay S. Horstmann – Core Java Volume I – Fundamentals, Pearson Education.
- E. Balagurusamy – Programming with Java: A Primer, McGraw-Hill Education.

Websites

- https://www.oracle.com/java – Official Java Documentation
- https://www.geeksforgeeks.org – Java Programming Tutorials and Examples
- https://www.w3schools.com – Java Programming Basics
- https://www.javatpoint.com – Core Java Concepts and Examples
- https://docs.oracle.com/en/java/ – Java API Documentation

Online Resources

- Java Object-Oriented Programming – Oracle Java Tutorials
- Payroll Calculation Concepts – HR Industry Standards Reference